Multimodal Image Captioning using Generative, Discriminative, and Fusion-Based Models

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1 Introduction

Image captioning is a multimodal task combining computer vision and natural language processing. The objective is to generate meaningful textual descriptions for images.

In this project, we designed a two-stage system:

- Generative models (LSTM and Transformer)
- Discriminative model for reranking captions

The system first generates multiple captions and then selects the best one using a learned scoring mechanism.

2 Dataset and Subset Selection

2.1 Dataset Description

This project uses the Amazon Berkeley Objects (ABO) dataset, which contains a large collection of product images along with metadata and textual descriptions.

Each data sample consists of:

- A product image
- Product metadata (JSON format)
- Textual descriptions or titles

2.2 Subset Selection

A footwear-only subset was selected:

- Shoes ($\sim$3500 samples)
- Sandals ($\sim$600 samples)
- Boots ($\sim$400 samples)

2.3 Data Preprocessing

Steps included:

- Cleaning noisy captions
- Lowercasing
- Tokenization
- Vocabulary construction (237 tokens)

Captions were standardized into:

$$\text{gender} + \text{color} + \text{material} + \text{type}$$

2.4 Train–Test Split

- 80% training (3623 samples)
- 20% testing (906 samples)

2.5 Dataset Statistics

- Total clean samples: 4529
- Discriminative pairs: 9058
- Caption length peaks at $\sim$5 words

3 Generative Modeling Component (My Contribution)

The generative modeling component was the primary part of the project that I implemented. This stage is responsible for producing candidate captions from raw image features, and its performance directly influences the quality of the final caption selected by the discriminative model.

3.1 Generative Model Workflow

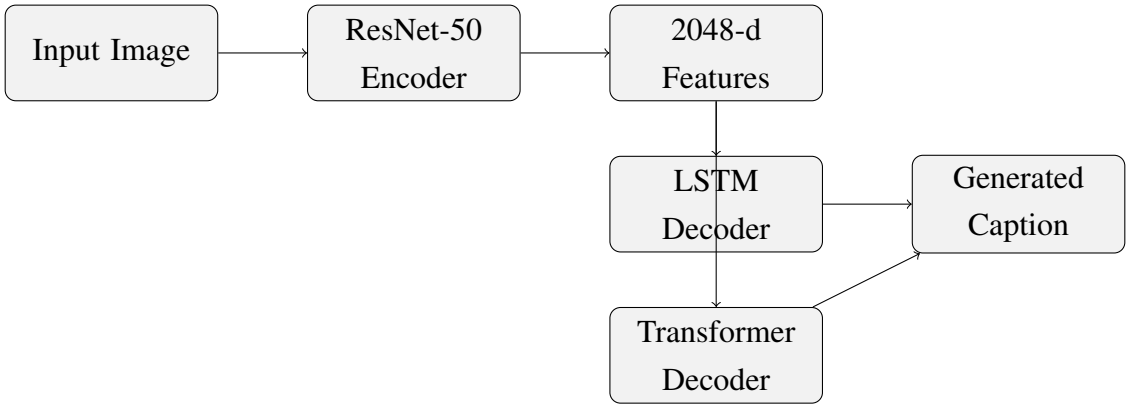


Figure 1: Workflow of the generative captioning models.

3.2 LSTM Captioning Model

Architecture

- Encoder: ResNet-50
- Embedding dimension: 256
- Hidden size: 512
- Vocabulary size: 237 tokens

Training Details

- Optimizer: Adam
- Learning rate: 1e-3
- Batch size: 32
- Epochs: 25
- Loss: Cross-entropy with padding mask

Epoch-wise Training (Initial 5 Epochs)

Epoch	Loss
1	3.2891
2	2.0509
3	1.6525
4	1.4080
5	1.2377

Continued Training (Checkpoint Loaded)

Epoch	Loss
1	1.1855
2	1.1007
3	1.0272
4	0.9708
5	0.9374
6	0.8863
7	0.8516
8	0.8179
9	0.7811
10	0.7562

Final LSTM Training

Epoch	Loss
1	1.7771
2	1.4064
3	1.3375
4	1.2782
5	1.2220
6	1.2542
7	1.2343
8	1.1961
9	1.1701
10	1.1429

3.3 Transformer Captioning Model

Architecture

- Encoder: ResNet-50 projected to model dimension
- Decoder: 2-layer Transformer
- Model dimension: 256
- Attention heads: 4
- Feed-forward dimension: 512

Training Details

- Optimizer: Adam with warmup
- Learning rate: 3e-4
- Batch size: 32
- Epochs: 30
- Label smoothing: 0.1

Final Transformer Training

Epoch	Loss
1	2.2900
2	1.6613
3	1.4630
4	1.3261
5	1.2376
6	1.1517
7	1.0875
8	1.0239
9	0.9755
10	0.9293

3.4 Overall Comparison of Generative Models

Model	Initial Loss	Final Loss	Behavior
LSTM	∼3.0	0.75–1.14	Stable, simple captions
Transformer	∼2.3	∼0.93	Detailed but less stable

3.5 Key Insights from Generative Model Training

- Both models show consistent loss reduction across epochs.
- LSTM converges faster and more smoothly, producing simpler but reliable captions.
- Transformer captures richer semantic details such as color, material, and shoe type.
- Transformer is more sensitive to dataset noise and requires careful masking.
- Incorrect early Transformer results highlight the importance of proper attention masking.
- Dataset quality and vocabulary size significantly influence caption diversity.

3.6 Training Loss Curve for Generative Models

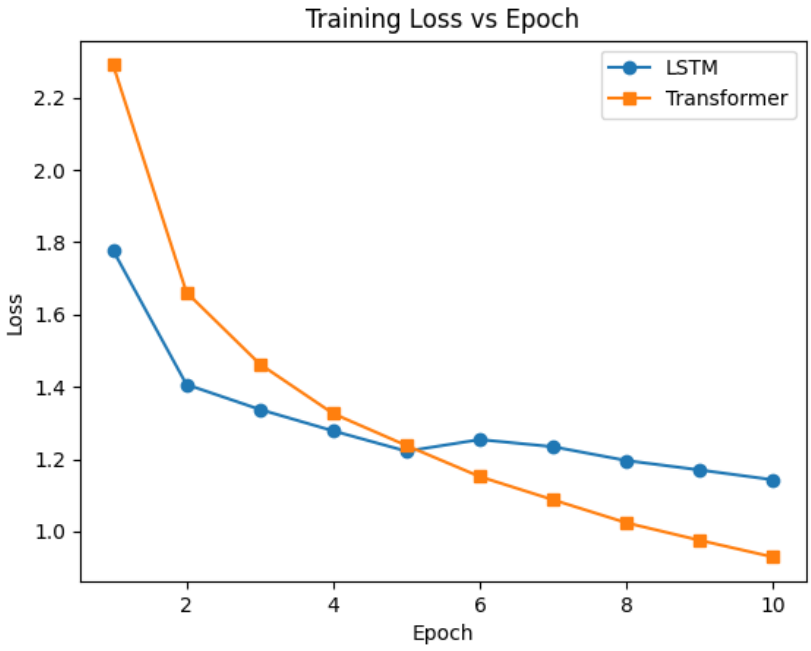


Figure 2: Training loss vs. epoch for LSTM and Transformer generative models.

3.7 Observations for Generative Models

LSTM Model Observations

- The LSTM model showed rapid loss reduction during the first few epochs, indicating fast initial learning.
- Training stabilized after approximately 5–7 epochs, with diminishing improvements in later epochs.
- Captions generated by the LSTM were generally shorter and more generic, often focusing on the shoe type rather than detailed attributes.
- The model performed reliably on common footwear categories but struggled with rare materials or unusual color combinations.
- Due to its sequential nature, the LSTM sometimes missed long-range dependencies, leading to incomplete attribute descriptions.

Transformer Model Observations

- The Transformer model converged more slowly but achieved lower final loss compared to the LSTM.
- Multi-head attention allowed the model to capture richer semantic details such as color, material, and shoe style.
- The model occasionally hallucinated attributes not present in the image, especially when trained without proper masking.
- Transformer performance was more sensitive to dataset noise and required careful preprocessing.
- Despite occasional errors, the Transformer consistently produced more descriptive and human-like captions.

3.8 Generative Model Evaluation

- BLEU-1: 0.4000
- BLEU-2: 0.2488
- BLEU-3: 0.1498

- BLEU-4: 0.1239
- ROUGE-L: 0.4876

4 Discriminative Model

4.1 Motivation

The discriminative model selects the best caption among multiple generated candidates.

4.2 Architecture

- Image encoder: ResNet-50
- Text encoder: BiLSTM
- Fusion: baseline, concatenation, gated
- Binary classifier

4.3 Performance

- Accuracy: 0.82
- AUC: 0.89

5 Fusion Models

5.1 Fusion Strategies

- Baseline fusion: AUC 0.4583
- Concatenation fusion: AUC 0.8194
- Gated fusion: AUC 0.8669

6 Results and Visualization

6.1 Generative Model Examples



Example 1



Example 2

Figure 3: Generative model examples (LSTM vs Transformer vs Ground Truth)

6.2 Discriminative Model Examples



Figure 4: Discriminative model prediction example 1

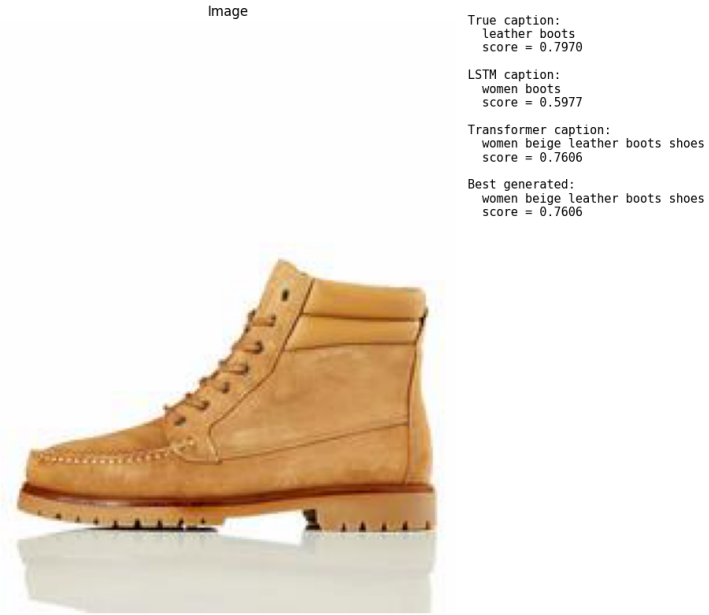


Figure 5: Discriminative model prediction example 2

6.3 Fusion Model Evaluation

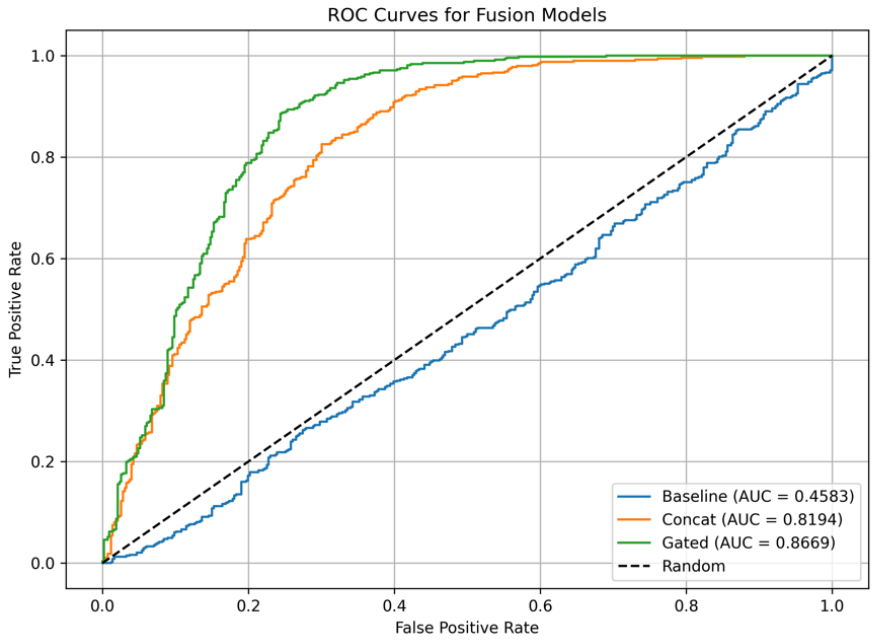


Figure 6: Fusion model ROC curves and performance metrics

7 Observations and Insights

- Transformer generates more detailed captions.
- LSTM produces simpler captions.
- Discriminator reliably selects the best caption.
- Gated fusion improves cross-modal alignment.

8 Challenges Faced

- Cleaning noisy captions
- Handling variable-length sequences
- Debugging Transformer decoding
- Domain mismatch issues

Motivation and Real-World Need

Image captioning has significant real-world applications across multiple industries. E-commerce platforms rely on accurate product descriptions for search indexing, recommendation systems, and accessibility. Automating caption generation reduces manual labeling effort and ensures consistent metadata quality across millions of product images.

Captioning systems also support accessibility tools for visually impaired users by providing descriptive text for images. In addition, digital asset management systems, content moderation pipelines, and multimodal AI applications benefit from reliable image-to-text models.

The two-stage design used in this project—generative captioning followed by discriminative reranking—mirrors real industry pipelines where one model generates candidates and another evaluates them. This approach improves caption accuracy and makes the system more robust in practical deployments.

9 Conclusion

This project integrates generative, discriminative, and fusion-based models into a unified multimodal captioning system. The Transformer produces richer captions, while the discriminative model improves final caption selection. The gated fusion model provides the strongest cross-modal alignment.